

KROONSTAD, South Africa

WMO: 683550

Lat: 27.666S

Lon: 27.314E

Elev: 1434

StdP: 85.24

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.0	1.4	-14.3	1.3	16.5	-12.5	1.5	13.8	11.1	17.4	9.4	14.3	2.6	100	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.3	33.1	15.3	31.7	15.2	30.5	15.2	20.1	24.9	19.0	25.0	18.4	24.4	4.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	16.2	21.8	17.5	14.9	20.6	16.7	14.2	20.4	64.7	24.1	60.4	24.9	58.4	24.4	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.9	8.8	7.8	DB	-2.3	36.0	2.0	2.1	-3.8	37.5	-5.0	38.7	-6.1	39.9	-7.6	41.5	(4)
(5)			WB	-4.0	20.2	1.5	1.0	-5.0	21.0	-5.9	21.6	-6.7	22.1	-7.7	22.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.9	23.1	22.0	20.9	17.9	14.3	11.1	10.7	13.8	17.7	20.3	21.2	22.7	(6)	
	DBStd	5.23	2.46	2.27	2.86	2.85	3.14	2.67	2.73	3.54	3.64	3.37	3.05	2.67	(7)	
	HDD10.0	64	0	0	0	1	8	19	26	9	1	0	0	0	(8)	
	HDD18.3	870	1	3	12	41	128	218	238	145	54	19	9	2	(9)	
	CDD10.0	2964	407	337	337	237	140	51	47	127	233	321	334	392	(10)	
	CDD18.3	728	150	107	90	27	2	0	0	4	36	81	94	136	(11)	
	CDH23.3	5784	1189	683	577	161	31	1	1	59	399	752	816	1116	(12)	
	CDH26.7	1887	438	192	150	24	1	0	0	4	112	260	285	422	(13)	
(14) Wind	WSAvg	3.7	4.2	3.7	3.4	3.0	2.8	3.0	3.1	3.7	4.1	4.4	4.6	4.5	(14)	
(15) Precipitation	PrecAvg	591	91	78	83	57	20	7	5	11	21	62	78	77	(15)	
	PrecMax	873	218	201	220	172	109	49	44	127	171	227	244	223	(16)	
	PrecMin	294	17	6	7	0	0	0	0	0	0	0	14	7	(17)	
	PrecStd	125	53	46	47	47	26	10	10	23	31	49	47	40	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.1	32.6	32.2	29.4	26.0	22.2	22.7	27.1	31.5	33.2	33.9	35.6	(19)	
		MCWB	16.2	16.1	16.1	14.9	12.1	9.7	9.5	11.6	12.8	13.9	14.3	15.4	(20)	
	2%	DB	32.6	30.8	30.0	26.8	24.0	20.5	20.8	24.9	29.7	31.4	31.8	32.7	(21)	
		MCWB	16.2	16.1	16.0	14.4	11.6	9.2	8.7	10.5	12.5	13.5	14.4	15.7	(22)	
	5%	DB	30.9	29.1	28.3	25.1	22.5	19.0	19.3	23.1	27.8	29.6	29.9	30.7	(23)	
		MCWB	16.8	16.6	16.1	14.1	11.4	8.7	8.1	9.8	11.7	13.1	14.3	15.9	(24)	
	10%	DB	29.2	27.6	26.6	23.5	20.9	17.4	17.5	21.1	25.7	27.6	28.0	28.9	(25)	
		MCWB	16.7	16.9	15.9	13.9	10.9	8.3	7.5	9.0	11.0	12.9	14.2	16.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.0	19.5	19.0	17.4	18.1	15.2	10.7	12.8	15.0	17.2	18.0	22.4	(27)	
		MCDB	27.6	25.1	24.6	23.6	20.3	16.9	19.1	23.7	25.6	24.5	24.7	24.7	(28)	
	2%	WB	19.5	19.0	18.2	16.5	14.6	11.8	9.6	11.7	13.9	16.1	17.2	21.1	(29)	
		MCDB	24.8	25.0	24.3	22.2	18.5	15.0	18.3	21.9	24.2	23.9	24.0	24.1	(30)	
	5%	WB	18.7	18.6	17.7	15.8	13.0	10.4	8.8	10.9	13.0	15.4	16.7	19.2	(31)	
		MCDB	24.5	24.5	23.9	21.2	18.7	16.2	17.4	20.6	23.8	23.3	23.5	25.9	(32)	
	10%	WB	18.2	18.1	17.1	15.1	12.0	9.2	8.0	9.9	12.2	14.7	16.2	17.9	(33)	
		MCDB	24.1	23.9	23.3	20.9	18.3	15.6	16.4	19.2	22.6	22.7	23.0	24.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	10.5	10.8	11.5	13.5	13.7	14.3	14.2	14.8	13.6	12.9	11.7	(35)	
		MCDBR	14.1	13.2	13.3	13.3	15.1	15.2	15.8	15.9	16.5	15.8	15.3	14.7	(36)	
	5% WB	MCWBR	3.7	3.8	3.8	4.3	6.0	7.0	7.1	6.7	6.1	5.0	4.5	4.0	(37)	
		MCWBR	9.7	9.9	10.6	10.3	12.2	12.5	14.4	14.2	14.7	12.7	11.8	10.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.368	0.369	0.355	0.314	0.283	0.268	0.267	0.322	0.362	0.388	0.344	0.353	(40)	
		taud	2.359	2.359	2.370	2.461	2.525	2.532	2.512	2.350	2.241	2.211	2.382	2.391	(41)	
	Ebn at Noon		974	956	938	932	922	917	931	908	915	926	991	991	(42)	
		Edh at Noon	133	129	121	100	84	79	84	110	135	148	129	129	(43)	
(44) All-Sky Solar Radiation	RadAvg		7.08	6.54	5.75	4.83	4.21	3.79	4.14	4.94	6.01	6.58	7.17	7.28	(44)	
	RadStd		0.67	0.58	0.42	0.37	0.19	0.17	0.13	0.25	0.27	0.44	0.48	0.42	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		+0.66	+0.75	N/A	N/A	N/A	N/A	-181	+234	+87

Nomenclature: See separate page